A function space characterization of RB-domains

Abstract

We prove that a domain D is an RB-domain if and only if the function space $[D \to E]$ is continuous for every domain E, thereby resolving a conjecture of Luan and Li. More precisely, for each domain D we construct an explicit algebraic test domain A_D such that continuity of the single function space $[D \to A_D]$ already forces D to be an RB-domain. For countably based domains D, the test domain can be chosen as a fixed domain that is independent of D. The analogous continuity characterization of FS-domains is false: the closed-disk domain is an FS-domain, but its function space into a suitable pointed algebraic domain is not continuous.

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1. Introduction

For domains D and E, the pointwise ordered set $[D \to E]$ of Scott-continuous maps is always a dcpo, but it need not be continuous even when both D and E are continuous. This is a basic obstruction to using all continuous domains as a higher-order semantic category [1, Section 4.3]. A central theme in domain theory is therefore to identify natural classes whose approximation structure is preserved by products and function spaces. Plotkin's bifinite domains give the classical algebraic example [15, 16]. In the continuous setting, Jung introduced FS-domains and proved their Cartesian closure [8, 9]; see also [1, Section 4.2]. RB-domains, equivalently the Scott-continuous retracts of bifinite domains, form a subclass of the FS-domains [18, Definition 2.2 and Lemma 2.4].

The main problem considered here is whether an RB-domain can be recognized from its outgoing function spaces. Miao, Luan, Liu and Li proved that $[D \to E]$ is continuous whenever D is an RB-domain and E is a domain [14]. Luan and Li asked whether the converse holds and gave a positive answer when D is algebraic, in which case RB-domains are precisely the bifinite domains [13, Section 4, Theorem 4.6]. The case of arbitrary continuous domains was left open. Our main result, stated and proved as Theorem 3.9, answers this question. It shows that, for every domain D,

$$D \in \mathbf{RB} \iff [D \to E] \text{ is continuous for every domain } E.$$

More precisely, from the order and the way-below relation of D we construct an explicit algebraic domain A_D such that continuity of the single function space $[D \to A_D]$ already implies that D is an RB-domain.

For countably based cases, the test target can be chosen independently of the source of the RB-domains. Theorem 3.13 constructs a fixed countably based algebraic domain A_ω such that, for every countably based domain D,

$$D \in \mathbf{RB} \iff [D \to A_\omega] \text{ is continuous.}$$

For pointed countably based domains, the same role is played by a fixed pointed countably based algebraic domain U_ω.

The analogous continuity statement for FS-domains is false, since the latest result shows that Disc, Lawson's planar closed-disk domain [1, 10] is an FS-domain but not an RB-domain [2]. Consequently, $[\mathsf{Disc} \to E_{\mathsf{Disc}}]$ is not continuous. We give a direct calculation showing how the pointed algebraic target E_{Disc} prevents $[\mathsf{Disc} \to E_{\mathsf{Disc}}]$ from being continuous. The proof combines the finite-support mechanism above with the finite-image Lorentz-flux theorem in [2, Proposition 5.3].

2. Preliminaries

We use standard domain-theoretic terminology as in [1, 4]. A *domain* is a nonempty continuous dcpo, and directed subsets are understood to be nonempty. For a set A, let $\mathrm{Fin}(A)$ denote the finite subsets of A, including $\varnothing$. We write

$\Uparrow x = \{y : x \ll y\}$, $\Downarrow x = \{y : y \ll x\}$. A subset B of a domain X is a basis if, for every $x \in X$, the set $B \cap \Downarrow x$ is directed with supremum x. An element k of a dcpo is compact if $k \ll k$. A dcpo is algebraic if every element is the supremum of the compact elements below it. A dcpo is pointed if it has a least element. A domain is countably based if it has a countable basis. For dcpos X and Y, the function space $[X \to Y]$ is the dcpo of Scott-continuous maps, ordered pointwise; directed suprema are computed pointwise. The notation $\ll_{\mathcal{O}(X)}$ denotes the way-below relation in the lattice of Scott-open subsets of X.

A directed family $\mathcal{F} \subseteq [X \to X]$ with $\bigvee \mathcal{F} = \mathrm{id}_X$ is called an approximate identity. Note that we use the unpointed convention and no idempotence condition is imposed on a deflation.

We use the following standard definitions; see [1].

Definition 2.1. Let X be a domain.

(a) A Scott-continuous map $f : X \to X$ is finitely separated from id_X if there is a finite set $M \subseteq X$ such that, for every $x \in X$, some $m \in M$ satisfies $f(x) \leq m \leq x$. The domain X is an *FS-domain* if it has an approximate identity consisting of maps finitely separated from the identity.

(b) A deflation on X is a Scott-continuous map $d : X \to X$ with finite image and $d \leq \mathrm{id}_X$. The domain X is an *RB-domain* if it has an approximate identity consisting of deflations.

Every deflation is finitely separated by its image, and hence **RB** $\subseteq$ **FS**. A pointed domain L is an *L-domain* if every principal ideal $\downarrow x$ is a complete lattice. A domain is *coherent* if the intersection of any two Scott-compact saturated sets is Scott compact. Every pointed domain is compact in its Scott topology. Consequently, for pointed domains, coherence is equivalent to Lawson compactness [1, Proposition 4.2.20]. We also use the following standard results.

Proposition 2.2. (i) *If D is an RB-domain and E is a domain, then $[D \to E]$ is continuous [14].*

(ii) *If D is a coherent domain and E is a pointed RB-domain, then $[D \to E]$ is an RB-domain [17, Corollary 4.9].*

(iii) *If D is a domain, E is a pointed domain, and $[D \to E]$ is continuous, then either D is coherent or E is an L-domain [1, Lemma 4.3.1].*

(iv) *If D and E are arbitrary domains and $[D \to E]$ is continuous, then either D is compact in its Scott topology or E is a disjoint union of pointed domains [1, Lemma 4.3.6].*

For a domain E, being a disjoint union of pointed domains is equivalent to every element of E lying above a unique minimal element.

3. Function-space characterization of RB-domains

Miao et al. [14] proved that if D is an RB-domain and E is any domain, then the function space $[D \to E]$ is continuous. Luan and Li [13] established the converse for algebraic domains and left open whether the algebraicity assumption can be removed. We will show that the converse is indeed true. Fix a domain X, we construct an algebraic test domain A_X. The continuity of $[X \to A_X]$ provides the finite approximation needed to construct finite-image deflations on X.

3.1. The algebraic test domain

Let $\mathcal{D}(X) = \{\varnothing\} \cup \{A \subseteq X : A \text{ is a directed subset of } X\}$ and let $\mathcal{P}(X)$ be the powerset of X. Define

$$E_X = (\mathcal{D}(X) \times \{0\}) \cup (\mathcal{P}(X) \times \{1\}),$$

ordered componentwise: $(A, i) \leq (B, j) \iff A \subseteq B$ and $i \leq j$. When the second coordinate is 0, the first coordinate must be directed or empty. When the second coordinate is 1, the first coordinate may be an arbitrary subset of X.

Lemma 3.1. *The poset E_X is a pointed algebraic domain. Its least and greatest elements are $(\varnothing, 0)$ and $(X, 1)$, respectively. The following compact elements form a basis:*

$$\mathcal{K}_X = \{(F, 0) : F = \varnothing \text{ or } F \subseteq X \text{ is finite and directed}\} \cup \{(M, 1) : M \in \mathrm{Fin}(X)\}.$$

Proof. Let $((A_i, \varepsilon_i))_{i\in I}$ be a directed family in E_X. If $\varepsilon_i = 0$ for every i, then $\bigcup_i A_i$ is empty or directed. Indeed, if $a \in A_i$ and $b \in A_j$, choose $k \in I$ above i and j. Then $a, b \in A_k$, and the directedness of A_k supplies $c \in A_k$ with $a, b \le c$. If some ε_i is 1, the first coordinate of the supremum need not be directed. In either case the supremum is

$$\bigvee_{i\in I}(A_i, \varepsilon_i) = \begin{cases} (\bigcup_i A_i, 0), & \varepsilon_i = 0 \text{ for every } i, \\ (\bigcup_i A_i, 1), & \varepsilon_i = 1 \text{ for some } i. \end{cases}$$

The same description also shows that $(\varnothing, 0)$ and $(X, 1)$ are, respectively, the least and greatest elements of E_X.

We next verify compactness. The least element $(\varnothing, 0)$ is compact. Suppose $(F, 0) \le \bigvee_i (A_i, \varepsilon_i)$, where F is nonempty, finite and directed. For every $a \in F$, choose i_a with $a \in A_{i_a}$; since F is finite and the indexing family is directed, one index i lies above all the i_a. Then $F \subseteq A_i$, and hence $(F, 0) \le (A_i, \varepsilon_i)$. If $(M, 1) \le \bigvee_i (A_i, \varepsilon_i)$, choose an index i_0 whose second coordinate is 1, and for each $m \in M$ choose i_m with $m \in A_{i_m}$. A common upper index i satisfies $M \subseteq A_i$ and $\varepsilon_i = 1$, so $(M, 1) \le (A_i, 1)$. Thus every element of $\mathcal{K}_X$ is compact.

Let $A \subseteq X$ be directed. Its finite directed subsets form a directed family under inclusion. Indeed, if F and G are two such subsets, directedness of A gives an element $a \in A$ above the finite set $F \cup G$, and $F \cup G \cup \{a\}$ is a finite directed subset of A containing both. Their union is A, and therefore $(A, 0) = \bigvee\{(F, 0) : F \subseteq A \text{ is finite and directed}\}$. The case $A = \varnothing$ is immediate. When the second coordinate is 1, the finite subsets of an arbitrary $A \subseteq X$ are directed by finite union and have union A, so $(A, 1) = \bigvee\{(M, 1) : M \subseteq A \text{ is finite}\}$. Hence every element is the supremum of compact elements below it, and E_X is algebraic. □

Define $Q : X \to E_X$ by $Q(x) = (\Downarrow x, 0)$. Thus the first coordinate of $Q(x)$ is the directed set of all way-below approximants of x.

Lemma 3.2. *The map Q is Scott continuous.*

Proof. If $x \le y$, then $\Downarrow x \subseteq \Downarrow y$, so Q is monotone. Let $D \subseteq X$ be directed. The inclusion $\bigcup_{x\in D} \Downarrow x \subseteq \Downarrow(\bigvee D)$ is immediate. Conversely, if $u \ll \bigvee D$, interpolation gives w with $u \ll w \ll \bigvee D$. Since $w \ll \bigvee D$, there is $x \in D$ with $w \le x$, and then $u \ll x$. Hence $\Downarrow(\bigvee D) = \bigcup_{x\in D} \Downarrow x$. All values $Q(x)$ have second coordinate 0, so the directed-supremum formula in Lemma 3.1 now gives $Q(\bigvee D) = \bigvee_{x\in D} Q(x)$. Thus Q is Scott continuous. □

We use the standard step function of Erker–Escardó–Keimel [3]. Let Y be a pointed dcpo, let k be compact in Y, let $f : X \to Y$ be Scott continuous, and let V be Scott open. Define

$$s_{V,k}(x) = \begin{cases} k, & x \in V, \\ \bot_Y, & x \notin V. \end{cases}$$

Since V is Scott open, the map $s_{V,k}$ is Scott continuous.

Lemma 3.3. *Let X be a dcpo and Y a pointed dcpo. Let $V \in \mathcal{O}(X)$, let k be a compact element of Y, and let $f \in [X \to Y]$. If $V \ll_{\mathcal{O}(X)} f^{-1}(\uparrow k)$, then $s_{V,k} \ll f$ in $[X \to Y]$.*

Proof. Let $(g_i)_{i\in I}$ be a directed family in $[X \to Y]$ such that $f \le \bigvee_{i\in I} g_i$. For each $i \in I$, put $W_i = g_i^{-1}(\uparrow k)$. Since k is compact, $\uparrow k$ is Scott open, so each W_i is Scott open. Moreover, the family $(W_i)_{i\in I}$ is directed under inclusion.

We claim that $f^{-1}(\uparrow k) \subseteq \bigcup_{i\in I} W_i$. Indeed, if $x \in f^{-1}(\uparrow k)$, then $k \le f(x) \le \bigvee_{i\in I} g_i(x)$. Since k is compact and $(g_i(x))_{i\in I}$ is directed, there exists $i \in I$ such that $k \le g_i(x)$. Hence $x \in W_i$.

Because $V \ll_{\mathcal{O}(X)} f^{-1}(\uparrow k)$, there exists $i \in I$ such that $V \subseteq W_i$. Thus $s_{V,k}(x) \le g_i(x)$ for every $x \in X$: if $x \in V$, then $k \le g_i(x)$, while if $x \notin V$, then $\bot_Y \le g_i(x)$. Therefore $s_{V,k} \le g_i$, and hence $s_{V,k} \ll f$. □

For $u \ll v$ in X, we have $\Uparrow v \subseteq \uparrow v \subseteq \Uparrow u$. The principal upset $\uparrow v$ is Scott compact. Hence every directed Scott-open cover whose union contains $\Uparrow u$ has one member containing $\uparrow v$, and therefore one member containing $\Uparrow v$. Thus $\Uparrow v \ll_{\mathcal{O}(X)} \Uparrow u$. Define

$$S_{u,v}(x) = \begin{cases} (\{u\}, 0), & v \ll x, \\ (\varnothing, 0), & v \not\ll x. \end{cases}$$

The element $(\{u\}, 0)$ is compact in E_X, and $Q^{-1}(\uparrow(\{u\}, 0)) = \Uparrow u$. This yields the following consequence.

Lemma 3.4. *Let X be a domain, and let E_X, $Q : X \to E_X$, and $S_{u,v} : X \to E_X$ be defined as above. If $u, v \in X$ satisfy $u \ll v$, then $S_{u,v} \ll Q$ in the function space $[X \to E_X]$.*

Proof. We have $\Uparrow v \ll_{O(X)} \Uparrow u$ and $Q^{-1}(\uparrow(\{u\}, 0)) = \Uparrow u$. Since $(\{u\}, 0)$ is compact in E_X, Lemma 3.3 applied to $V = \Uparrow v$, $k = (\{u\}, 0)$ and $f = Q$ yields $S_{u,v} \ll Q$. □

For $M \in \mathrm{Fin}(X)$, let $K_M : X \to E_X$ be the constant function with value $(M, 1)$. The family $(K_M)_{M \in \mathrm{Fin}(X)}$ is directed and $\bigvee_{M \in \mathrm{Fin}(X)} K_M$ is the constant function with value $(X, 1)$. Since Q lies below this constant top function, every $\varphi \ll Q$ satisfies $\varphi \leq K_M$ for some finite $M \subseteq X$. The following lemma turns this finite directed information into a deflation.

Lemma 3.5. *Let X be a domain, let M be a finite subset of X, and let $\varphi : X \to E_X$ be Scott continuous. Suppose that $\varphi \leq Q$ and $\varphi \leq K_M$. For each $x \in X$, write $\varphi(x) = (B_\varphi(x), 0)$. Then $B_\varphi(x) \in \mathcal{D}(X)$ and $B_\varphi(x) \subseteq M \cap \Downarrow x$. If $B_\varphi(x)$ is nonempty for every $x \in X$, then $d_\varphi(x) = \max B_\varphi(x)$ defines a deflation on X whose image is contained in M. Moreover, if $\psi : X \to E_X$ satisfies the same assumptions and $\varphi \leq \psi$, then $d_\varphi \leq d_\psi$.*

Proof. Since $\varphi(x) \leq Q(x) = (\Downarrow x, 0)$, the second coordinate of $\varphi(x)$ is 0 and $B_\varphi(x) \subseteq \Downarrow x$. The inequality $\varphi(x) \leq (M, 1)$ gives $B_\varphi(x) \subseteq M$. This proves the stated inclusions. If $B_\varphi(x)$ is nonempty, then it is a finite directed set and therefore has a greatest element. Hence d_φ is well defined, has image contained in M, and satisfies $d_\varphi(x) \in B_\varphi(x) \subseteq \Downarrow x$. In particular $d_\varphi(x) \ll x$, so $d_\varphi \leq \mathrm{id}_X$.

The map d_φ is monotone. Indeed, $x \leq y$ and the monotonicity of φ imply $B_\varphi(x) \subseteq B_\varphi(y)$, and hence $d_\varphi(x) \leq d_\varphi(y)$. To prove Scott continuity, let $D \subseteq X$ be directed and put $x = \bigvee D$. Since φ is Scott continuous and all its values have second coordinate 0, $B_\varphi(x) = \bigcup_{y \in D} B_\varphi(y)$. The element $d_\varphi(x)$ belongs to this union, so it belongs to $B_\varphi(y_0)$ for some $y_0 \in D$. Since $d_\varphi(x) \in B_\varphi(y_0)$ and $d_\varphi(y_0)$ is the greatest element of $B_\varphi(y_0)$, we have $d_\varphi(x) \leq d_\varphi(y_0)$. Monotonicity and $y_0 \leq x$ give the reverse inequality. Hence $d_\varphi(y_0) = d_\varphi(x)$, and monotonicity then gives $d_\varphi(y) = d_\varphi(x)$ for every $y \in D$ above y_0. It follows that $d_\varphi(\bigvee D) = \bigvee_{y \in D} d_\varphi(y)$. Consequently d_φ is a deflation.

Finally, if $\varphi \leq \psi$, then $B_\varphi(x) \subseteq B_\psi(x)$ for every x. Taking greatest elements gives $d_\varphi(x) \leq d_\psi(x)$, and hence $d_\varphi \leq d_\psi$. □

Proposition 3.6. *Let X be a domain. Suppose that $[X \to E_X]$ is continuous and that there are finitely many pairs $u_i \ll v_i$ $(1 \leq i \leq m)$ such that $X = \Uparrow v_1 \cup \cdots \cup \Uparrow v_m$. Then X is an RB-domain.*

Proof. Because $[X \to E_X]$ is continuous, $\Downarrow Q$ is directed and has supremum Q. By Lemma 3.4, each S_{u_i,v_i} belongs to $\Downarrow Q$, so directedness gives some $\varphi_0 \ll Q$ such that $S_{u_i,v_i} \leq \varphi_0$ for all $1 \leq i \leq m$. Let $\mathcal{D} = \{\varphi \ll Q : \varphi_0 \leq \varphi\}$. The set $\mathcal{D}$ is directed. It is also cofinal in $\Downarrow Q$: given $\psi \ll Q$, directedness of $\Downarrow Q$ supplies $\varphi \ll Q$ above both ψ and φ_0.

Fix $\varphi \in \mathcal{D}$. Since $\varphi \ll Q$, one has $\varphi \leq Q$. Since $Q \leq \bigvee_{M \in \mathrm{Fin}(X)} K_M$, there exists a finite $M \subseteq X$ with $\varphi \leq K_M$. Write $\varphi(x) = (B_\varphi(x), 0)$. For each $x \in X$, choose i with $v_i \ll x$. Then $(\{u_i\}, 0) = S_{u_i,v_i}(x) \leq \varphi_0(x) \leq \varphi(x)$, so $u_i \in B_\varphi(x)$. Thus every $B_\varphi(x)$ is nonempty, and Lemma 3.5 yields a deflation d_φ. If $\varphi, \psi \in \mathcal{D}$, choose $\chi \in \mathcal{D}$ above both. The last assertion of Lemma 3.5 gives $d_\varphi, d_\psi \leq d_\chi$. Hence $(d_\varphi)_{\varphi \in \mathcal{D}}$ is directed.

Since all d_φ are all deflations, it remains to compute their pointwise supremum. Let $x \in X$ and $u \ll x$. By interpolation, choose $v \in X$ such that $u \ll v \ll x$. Since $S_{u,v} \ll Q$ and the family $\mathcal{D}$ is cofinal below Q, there exists $\varphi \in \mathcal{D}$ with $S_{u,v} \leq \varphi$. Evaluating at x gives $(\{u\}, 0) = S_{u,v}(x) \leq \varphi(x) = (B_\varphi(x), 0)$. Hence $u \in B_\varphi(x)$, and therefore $u \leq d_\varphi(x) \leq \bigvee_{\psi \in \mathcal{D}} d_\psi(x)$. Since this holds for every $u \ll x$ and X is continuous, $x = \bigvee \Downarrow x \leq \bigvee_{\psi \in \mathcal{D}} d_\psi(x)$. The reverse inequality follows from $d_\varphi \leq \mathrm{id}_X$ for every φ. Thus $\bigvee_{\varphi \in \mathcal{D}} d_\varphi = \mathrm{id}_X$, so X is an RB-domain. □

Corollary 3.7. *If X is a pointed domain, then $[X \to E_X]$ is continuous if and only if X is an RB-domain.*

Proof. The forward implication follows from Proposition 3.6 with the single pair $u = v = \bot_X$, since $\bot_X \ll \bot_X$ and $\Uparrow \bot_X = X$. The reverse implication is Proposition 2.2 (i). □

In the following, we will use E_X combined with a three-point domain to construct the final test domain. Let $V = \{\ell, r, \top\}$, ordered by $\ell < \top$ and $r < \top$, with ℓ and r incomparable. The finite poset V is an algebraic domain. Its two minimal elements are ℓ and r, while $\top$ lies above both. Consequently, V is not a disjoint union of pointed domains.

Define $A_X = E_X \times V$. Since finite products of algebraic dcpos are algebraic, Lemma 3.1 shows that A_X is an algebraic domain. It is not a disjoint union of pointed domains: the two distinct minimal elements $((\varnothing, 0), \ell)$ and $((\varnothing, 0), r)$ both lie below $((\varnothing, 0), \top)$.

Theorem 3.8. *Let X be a domain. If $[X \to A_X]$ is continuous, then X is an RB-domain.*

Proof. Let $p_E : A_X \to E_X$ be the first-coordinate projection and define $s_E : E_X \longrightarrow A_X$, $s_E(e) = (e, \ell)$. Then $p_E \circ s_E = \mathrm{id}_{E_X}$. Define

$$S : [X \to E_X] \longrightarrow [X \to A_X], \quad S(f) = s_E \circ f,$$

and

$$P : [X \to A_X] \longrightarrow [X \to E_X], \quad P(g) = p_E \circ g.$$

Both maps are Scott continuous because directed suprema in the two function spaces are computed pointwise and s_E and p_E preserve directed suprema. Since $p_E \circ s_E = \mathrm{id}_{E_X}$, for every $f \in [X \to E_X]$ we have

$$(P \circ S)(f) = p_E \circ s_E \circ f = f.$$

Thus $P \circ S = \mathrm{id}_{[X \to E_X]}$, so $[X \to E_X]$ is a Scott-continuous retract of $[X \to A_X]$. Since $[X \to A_X]$ is continuous and Scott-continuous retracts of continuous dcpos are continuous, $[X \to E_X]$ is continuous. By Proposition 2.2 (iv), the continuity of $[X \to A_X]$ implies that either X is compact in its Scott topology or A_X is a disjoint union of pointed domains. The latter alternative has already been ruled out. Therefore X is Scott compact. For each $x \in X$, continuity and interpolation provide elements $u \ll v \ll x$. Thus the Scott-open sets $\Uparrow v$ ($u \ll v$ for some $u \in X$) cover X. Scott compactness yields finitely many pairs $u_i \ll v_i$ such that $X = \Uparrow v_1 \cup \cdots \cup \Uparrow v_m$. Proposition 3.6, applied to the already established continuity of $[X \to E_X]$, now shows that X is an RB-domain. □

Theorem 3.9. *For every domain X, the following conditions are equivalent:*

(i) *X is an RB-domain;*

(ii) *$[X \to E]$ is continuous for every domain E.*

Moreover, the algebraic domain A_X constructed above has the property that continuity of $[X \to A_X]$ already implies that X is an RB-domain.

Proof. The implication (i)⇒(ii) is the theorem of Miao–Luan–Liu–Li [14]. Conversely, assume (ii). The algebraic domain A_X is a domain, so $[X \to A_X]$ is continuous; Theorem 3.8 then yields (i). The same theorem gives the final, single-target assertion. □

3.2. A uniform test domain for countably based domains

The target A_X is built from the underlying set and order of X. For countably based domains, this dependence can be removed by placing all finite order patterns into one countable algebraic domain.

For a poset P, let $\mathrm{Idl}(P)$ denote the dcpo of nonempty directed lower subsets of P, ordered by inclusion. A *finite rooted order* on $\mathbb{N}$ is a triple $\alpha = (F_\alpha, \preccurlyeq_\alpha, t_\alpha)$, where F_α is a nonempty finite subset of $\mathbb{N}$, $\preccurlyeq_\alpha$ is a partial order on F_α, and t_α is its greatest element. Let $\mathcal{R}_\omega$ be the set of all finite rooted orders on $\mathbb{N}$. Define a poset P_ω whose elements are $\bot_P$, the members of $\mathcal{R}_\omega$, and symbols u_M for $M \in \mathrm{Fin}(\mathbb{N})$. The element $\bot_P$ is least, and the remaining order relations are

$$\alpha \le \beta \iff F_\alpha \subseteq F_\beta \text{ and } \preccurlyeq_\alpha \subseteq \preccurlyeq_\beta,$$

$$\alpha \le u_M \iff F_\alpha \subseteq M, \qquad u_M \le u_N \iff M \subseteq N.$$

There are no other comparabilities.

Lemma 3.10. *The poset P_ω is countable and directed. Its ideal completion $U_\omega = \mathrm{Idl}(P_\omega)$ is a pointed countably based algebraic domain.*

Proof. The displayed clauses define a partial order. For example, if $\alpha \le \beta \le u_M$, then $F_\alpha \subseteq F_\beta \subseteq M$, so $\alpha \le u_M$. Antisymmetry among rooted orders follows because a finite partial order has at most one greatest element. Every finite subset of P_ω has an upper bound u_M, where M is the union of all supports and all finite sets occurring in that subset. Hence P_ω is directed. It is countable because it is built from finite subsets and finite relations on $\mathbb{N}$. The principal ideals form a countable compact basis of its ideal completion, which is pointed because P_ω has a least element. □

Let X be a countably based domain. Choose a countable basis $B = \{b_n : n \in L\}$, where $L \subseteq \mathbb{N}$ and the elements b_n are pairwise distinct. A rooted order $\alpha = (F_\alpha, \preccurlyeq_\alpha, t_\alpha)$ is called B-admissible if $F_\alpha \subseteq L$ and $n \preccurlyeq_\alpha m \implies b_n \leq b_m$ for all $n, m \in F_\alpha$. For each $x \in X$, define

$$Q_B(x) = \{\bot_P\} \cup \{\alpha \in \mathcal{R}_\omega : \alpha \text{ is } B\text{-admissible and } b_{t_\alpha} \ll x\}.$$

For $p \in P_\omega$, write $\widehat{p} = {\downarrow_{P_\omega}} p = \{q \in P_\omega : q \leq p\}$. Then $\widehat{p}$ is the principal ideal generated by p, and hence a compact element of $U_\omega = \mathrm{Idl}(P_\omega)$.

Lemma 3.11. *Let X be a countably based domain. For every $x \in X$, the set $Q_B(x)$ is an ideal of P_ω, and $Q_B : X \to U_\omega$ is Scott continuous.*

Proof. Suppose that $\alpha \leq \beta$ and $\beta \in Q_B(x)$. Then α is B-admissible. Since t_β is greatest in F_β and $t_\alpha \in F_\beta$, one has $b_{t_\alpha} \leq b_{t_\beta} \ll x$, so $\alpha \in Q_B(x)$. Thus $Q_B(x)$ is lower.

Let $\alpha, \beta \in Q_B(x)$. Since $B \cap \Downarrow x$ is directed, there is $b_r \in B \cap \Downarrow x$ above both b_{t_α} and b_{t_β}. On $F_\alpha \cup F_\beta \cup \{r\}$ define $n \preccurlyeq m \iff b_n \leq_X b_m$. Then r is the greatest element of this finite order. With r as greatest element, this gives a B-admissible rooted order $\gamma \in Q_B(x)$ above α and β. Hence $Q_B(x)$ is directed.

The map Q_B is monotone. Moreover, for a rooted order α, we have

$$Q_B^{-1}(\uparrow\widehat{\alpha}) = \begin{cases} \Uparrow b_{t_\alpha}, & \alpha \text{ is } B\text{-admissible}, \\ \varnothing, & \text{otherwise}, \end{cases}$$

while $Q_B^{-1}(\uparrow\widehat{u_M}) = \varnothing$ and $Q_B^{-1}(\uparrow\widehat{\bot_P}) = X$. These are Scott open, and the principal ideals form a compact basis of U_ω. Therefore Q_B is Scott continuous. □

For $M \in \mathrm{Fin}(\mathbb{N})$, let $K_M : X \to U_\omega$ be constant with value $\widehat{u_M}$. The maps K_M form a directed family whose supremum is the constant top map. Consequently, every $\varphi \ll Q_B$ satisfies $\varphi \leq K_M$ for some finite M. If $\varphi \leq Q_B$, put

$$L_\varphi(x) = \{b_{t_\alpha} : \alpha \in \varphi(x) \cap \mathcal{R}_\omega\}.$$

Every rooted order occurring in $\varphi(x)$ has support contained in M, so $L_\varphi(x)$ is a finite subset of $\{b_n : n \in M \cap L\} \cap \Downarrow x$. It is directed: a common upper bound of two rooted orders in the ideal $\varphi(x) \subseteq Q_B(x)$ is again a rooted order, and its greatest label gives an upper bound of the two corresponding basis elements. Whenever $L_\varphi(x)$ is nonempty, it therefore has a greatest element. The proof of Lemma 3.5, with $L_\varphi(x)$ in place of $B_\varphi(x)$, shows that $d_\varphi(x) = \max L_\varphi(x)$ is a finite-image deflation. It also shows that $\varphi \leq \psi$ implies $d_\varphi \leq d_\psi$ whenever both maps are defined.

For $n \in L$, let α_n be the rooted order on the singleton $\{n\}$. If $b_n \ll v$, define $S_{n,v} : X \to U_\omega$ by

$$S_{n,v}(x) = \begin{cases} \widehat{\alpha_n}, & v \ll x, \\ \widehat{\bot_P}, & v \not\ll x. \end{cases}$$

Since $Q_B^{-1}(\uparrow\widehat{\alpha_n}) = \Uparrow b_n$ and $\Uparrow v \ll_{O(X)} \Uparrow b_n$, Lemma 3.3 gives $S_{n,v} \ll Q_B$.

Proposition 3.12. *Let X be a countably based domain. Suppose that $[X \to U_\omega]$ is continuous and that $X = \Uparrow v_1 \cup \cdots \cup \Uparrow v_m$ for some $b_{n_i} \ll v_i$, where $n_i \in L$. Then X is an RB-domain.*

Proof. Choose $\varphi_0 \ll Q_B$ above the finitely many maps S_{n_i,v_i}, and let $\mathcal{D}$ be the set of all $\varphi \ll Q_B$ above φ_0. This is a directed cofinal subset of $\Downarrow Q_B$. For every $\varphi \in \mathcal{D}$, the finite bound above and the covering condition make each $L_\varphi(x)$ nonempty, so d_φ is a finite-image deflation. These deflations form a directed family.

Let $u \ll x$. By the basis property and interpolation, choose $n \in L$ and $v \in X$ with $u \leq b_n \ll v \ll x$. Cofinality gives $\varphi \in \mathcal{D}$ with $S_{n,v} \leq \varphi$, and then $u \leq b_n \leq d_\varphi(x)$. For every $u \ll x$, some φ satisfies $u \leq d_\varphi(x)$. Since X is continuous, $x = \bigvee \Downarrow x \leq \bigvee_\varphi d_\varphi(x)$. The reverse inequality follows from $d_\varphi \leq \mathrm{id}_X$. Hence X is an RB-domain. □

Let V be the three-element algebraic domain used in the preceding subsection and put $A_\omega = U_\omega \times V$. This is a countably based algebraic domain. It is not a disjoint union of pointed domains, because its two minimal elements $(\widehat{\bot_P}, \ell)$ and $(\widehat{\bot_P}, r)$ have the common upper bound $(\widehat{\bot_P}, \top)$.

Theorem 3.13. *For every countably based domain X,*

$$X \in \mathbf{RB} \iff [X \to A_\omega] \text{ is continuous.}$$

If X is pointed, then

$$X \in \mathbf{RB} \iff [X \to U_\omega] \text{ is continuous.}$$

Proof. If X is an RB-domain, both function spaces are continuous by Proposition 2.2 (i). Conversely, suppose that $[X \to A_\omega]$ is continuous. Projection onto the first coordinate and the section $e \mapsto (e, \ell)$ show, exactly as in the proof of Theorem 3.8, that $[X \to U_\omega]$ is a Scott-continuous retract of $[X \to A_\omega]$, and hence is continuous. Proposition 2.2 (iv) implies that X is Scott compact, because A_ω is not a disjoint union of pointed domains.

For every $x \in X$, choose $b_n \in B \cap \Downarrow x$ and then choose v with $b_n \ll v \ll x$. The corresponding sets $\Uparrow v$ cover X, so Scott compactness gives a finite subcover. Proposition 3.12 now yields $X \in \mathbf{RB}$.

If X is pointed, enlarge the chosen basis by $\perp_X$ if necessary. The single choice $b_n = v = \perp_X$ satisfies $b_n \ll v$ and $\Uparrow v = X$. Proposition 3.12 therefore gives the converse from the continuity of $[X \to U_\omega]$. □

4. Function spaces of FS-domains

The RB characterization has two natural FS analogues: one may ask whether an FS-domain preserves continuity of the function space for all targets, or whether FS-domains can be detected from the class of function spaces it generates. Of course, if FS-domains are not identical to RB-domains, then the first analogue is false. We will use an example to show that. The second becomes correct after strengthening the conclusion from continuity to FS-membership.

4.1. Failure of the continuity analogue

Let Disc be the dcpo of all closed disks in the Euclidean plane, together with the whole plane as its least element, ordered by reverse inclusion. Its FS property is recorded in [1, Section 4.2.2, example following Proposition 4.2.12]; it also follows from Lawson's formal-ball theorem [10]. Using [2, Proposition 5.3], we give a direct function-space obstruction showing that $[\mathsf{Disc} \to E_{\mathsf{Disc}}]$ is not continuous.

Represent a non-bottom disk $\overline{B}(\mathbf{z}, r)$ by $\mathbf{x} = (-r, \mathbf{z}) \in H := (-\infty, 0] \times \mathbb{R}^2$. Throughout this subsection, $\|\cdot\|$ denotes the Euclidean norm and $\mathbf{v} \otimes \mathbf{w}$ denotes the matrix $\mathbf{v}\mathbf{w}^T$. Let $C = \{(t, \mathbf{z}) \in \mathbb{R} \times \mathbb{R}^2 : t \geq \|\mathbf{z}\|\}$ be the three-dimensional Lorentz cone, and write $\mathbf{x} \leq_C \mathbf{y}$ when $\mathbf{y} - \mathbf{x} \in C$. A map between subsets of $\mathbb{R}^3$ is called C-monotone if it is order preserving with respect to $\leq_C$. Under the representation above, reverse inclusion of disks is precisely the order $\leq_C$, and for non-bottom elements $\mathbf{x} \ll \mathbf{y} \iff \mathbf{y} - \mathbf{x} \in \operatorname{int} C$ [2, Section 3]. Put $J = \operatorname{diag}(1, -1, -1)$ and $\Lambda(A) = \operatorname{tr}(JA)$. If $\mathbf{v} = (v_0, \mathbf{v}')$ and $\mathbf{w} = (w_0, \mathbf{w}')$ belong to C, then

$$\Lambda(\mathbf{v} \otimes \mathbf{w}) = \mathbf{w}^T J \mathbf{v} = v_0 w_0 - \mathbf{v}' \cdot \mathbf{w}' \geq 0,$$

whereas $\Lambda(I_3) = -1$. We use the following scalar consequence of [2, Proposition 5.3].

Proposition 4.1. *Let $\Omega \subseteq \mathbb{R}^3$ be an open rectangular box, let $q : \Omega \to \mathbb{R}^3$ be a finite-image C-monotone map, and let $0 \leq \psi \in C_c^\infty(\Omega)$. Then q is Lebesgue measurable, the following matrix integral is well defined coordinatewise, and*

$$\Lambda\left(-\int_\Omega q(\mathbf{x}) \otimes \nabla\psi(\mathbf{x})\, d\mathbf{x}\right) \geq 0.$$

Proof. Proposition [2, Proposition 5.3] gives measurability and places the tested matrix in the closed convex cone generated by $\mathbf{v} \otimes \mathbf{w}$ with $\mathbf{v}, \mathbf{w} \in C$. The preceding calculation shows that Λ is nonnegative on every generator and hence on that cone. □

Fix $P = [-6, -4] \times [-1, 1]^2$, $\Omega = (-6, -4) \times (-1, 1)^2$, and choose $0 \leq \psi \in C_c^\infty(\Omega)$ with $\int_\Omega \psi = 1$. Set $L_\psi = \int_\Omega \|\nabla\psi(\mathbf{x})\|\, d\mathbf{x}$ and choose $\eta > 0$ such that $\eta L_\psi < 1$. Let $\mathbf{e} = (1, 0, 0)$.

Lemma 4.2. *If $\mathbf{x} - \eta\mathbf{e} \leq_C \mathbf{y} \leq_C \mathbf{x}$, then $\|\mathbf{x} - \mathbf{y}\| \leq \eta$.*

Proof. Write $\mathbf{d} = \mathbf{x} - \mathbf{y} = (d_0, \mathbf{d}')$. The two order inequalities say that $\mathbf{d} \in C$ and $\eta\mathbf{e} - \mathbf{d} \in C$. Hence $d_0 \geq \|\mathbf{d}'\|$ and $\eta - d_0 \geq \|\mathbf{d}'\|$. Thus $0 \leq d_0 \leq \eta$ and $\|\mathbf{d}'\| \leq \min\{d_0, \eta - d_0\}$. If $d_0 \leq \eta/2$, then $\|\mathbf{d}\|^2 \leq 2d_0^2 \leq \eta^2/2$. If $d_0 \geq \eta/2$, then $\|\mathbf{d}\|^2 \leq d_0^2 + (\eta - d_0)^2 \leq \eta^2$. In either case $\|\mathbf{d}\| \leq \eta$. □

Proposition 4.3. *There is no finite-image C-monotone map $s : P \to H$ satisfying $\mathbf{x} - \eta\mathbf{e} \leq_C s(\mathbf{x}) \leq_C \mathbf{x}$ $(\mathbf{x} \in P)$.*

Proof. Suppose that such an s exists and put $q = s|_\Omega$. By Proposition 4.1, $\Lambda\Big(-\int_\Omega q \otimes \nabla\psi\Big) \geq 0$. Entrywise integration by parts gives $-\int_\Omega \mathbf{x} \otimes \nabla\psi(\mathbf{x})\, d\mathbf{x} = I_3$, because $-\int_\Omega x_i \partial_j \psi = \delta_{ij} \int_\Omega \psi = \delta_{ij}$. By Lemma 4.2, $\sup_\Omega \|q(\mathbf{x}) - \mathbf{x}\| \leq \eta$. Since J only changes the signs of the last two coordinates, $\|J\mathbf{z}\| = \|\mathbf{z}\|$ for every $\mathbf{z} \in \mathbb{R}^3$. We have

$$\left|\Lambda\Big(-\int_\Omega q \otimes \nabla\psi\Big) - \Lambda(I_3)\right| = \left|\int_\Omega \nabla\psi(\mathbf{x})^T J(\mathbf{x} - q(\mathbf{x}))\, d\mathbf{x}\right| \leq \sup_\Omega \|q(\mathbf{x}) - \mathbf{x}\| \int_\Omega \|\nabla\psi(\mathbf{x})\|\, d\mathbf{x} \leq \eta L_\psi < 1.$$

Together with $\Lambda(I_3) = -1$, this implies $\Lambda\Big(-\int_\Omega q \otimes \nabla\psi\Big) < 0$, contradicting the nonnegativity established above. □

For each $\mathbf{x} \in P$, put $\mathbf{u}_\mathbf{x} = \mathbf{x} - \frac{3\eta}{4}\mathbf{e}$, $\mathbf{v}_\mathbf{x} = \mathbf{x} - \frac{\eta}{2}\mathbf{e}$. Then $\mathbf{u}_\mathbf{x} \ll \mathbf{v}_\mathbf{x} \ll \mathbf{x}$ by the characterization of the way-below relation above. Moreover, the set $N_\mathbf{x} = \{\mathbf{y} \in H : \mathbf{y} - \mathbf{v}_\mathbf{x} \in \operatorname{int} C,\ \mathbf{u}_\mathbf{x} - \mathbf{y} + \eta\mathbf{e} \in \operatorname{int} C\}$ is a Euclidean neighborhood of $\mathbf{x}$. Indeed, at $\mathbf{y} = \mathbf{x}$ the two displayed vectors are respectively $(\eta/2)\mathbf{e}$ and $(\eta/4)\mathbf{e}$, both of which belong to $\operatorname{int} C$. Compactness of P therefore gives $\mathbf{x}_1, \ldots, \mathbf{x}_m \in P$ such that $P \subseteq \bigcup_{j=1}^m N_{\mathbf{x}_j}$. Write $\mathbf{u}_j = \mathbf{u}_{\mathbf{x}_j}$, $\mathbf{v}_j = \mathbf{v}_{\mathbf{x}_j}$. Then, for every $\mathbf{y} \in P$, some j satisfies $\mathbf{v}_j \ll \mathbf{y}$, $\mathbf{y} - \eta\mathbf{e} \leq_C \mathbf{u}_j$. Let $Q_{\mathsf{Disc}} : \mathsf{Disc} \to E_{\mathsf{Disc}}$, $Q_{\mathsf{Disc}}(x) = (\Downarrow x, 0)$, be the canonical map introduced above. For $j = 1, \ldots, m$, let $S_j = S_{\mathbf{u}_j, \mathbf{v}_j}$, where $S_{u,v}$ is the step map defined above. Since $\mathbf{u}_j \ll \mathbf{v}_j$, Lemma 3.4 gives $S_j \ll Q_{\mathsf{Disc}}$ $(1 \leq j \leq m)$.

Theorem 4.4. *The finite family $S_1, \ldots, S_m$ has no common upper bound in $\Downarrow Q_{\mathsf{Disc}}$. Consequently, $[\mathsf{Disc} \to E_{\mathsf{Disc}}]$ is not continuous.*

Proof. Assume, to the contrary, that there is $\varphi \ll Q_{\mathsf{Disc}}$ satisfying $S_j \leq \varphi \quad (1 \leq j \leq m)$. Since the constant maps K_M, $M \in \mathrm{Fin}(\mathsf{Disc})$, form a directed family whose supremum is the constant top map, and since Q_{Disc} lies below that supremum, the relation $\varphi \ll Q_{\mathsf{Disc}}$ yields a finite set $M \subseteq \mathsf{Disc}$ such that $\varphi \leq K_M$. Also $\varphi \leq Q_{\mathsf{Disc}}$. Hence, for every $x \in \mathsf{Disc}$, there is $B_\varphi(x) \in \mathcal{D}(\mathsf{Disc})$ such that $\varphi(x) = (B_\varphi(x), 0)$, $B_\varphi(x) \subseteq M \cap \Downarrow x$. Fix $\mathbf{y} \in P$ and choose j so that $\mathbf{v}_j \ll \mathbf{y}$ and $\mathbf{y} - \eta\mathbf{e} \leq_C \mathbf{u}_j$. Since $\mathbf{v}_j \ll \mathbf{y}$, the definition of S_j and the assumption $S_j \leq \varphi$ give $(\{\mathbf{u}_j\}, 0) = S_j(\mathbf{y}) \leq \varphi(\mathbf{y})$. Thus $\mathbf{u}_j \in B_\varphi(\mathbf{y})$. In particular, $B_\varphi(\mathbf{y})$ is a nonempty finite directed set. It therefore has a greatest element; define $s(\mathbf{y}) = \max B_\varphi(\mathbf{y})$ $(\mathbf{y} \in P)$. Since $B_\varphi(\mathbf{y})$ contains the non-bottom element $\mathbf{u}_j$, its greatest element is non-bottom and hence belongs to H. The image of s is contained in the finite set M. If $\mathbf{x} \leq_C \mathbf{y}$ in P, monotonicity of φ gives $B_\varphi(\mathbf{x}) \subseteq B_\varphi(\mathbf{y})$, and hence $s(\mathbf{x}) \leq_C s(\mathbf{y})$. Thus s is C-monotone.

For the lower estimate, choose j so that $\mathbf{v}_j \ll \mathbf{y}$ and $\mathbf{y} - \eta\mathbf{e} \leq_C \mathbf{u}_j$. Since $\mathbf{u}_j \in B_\varphi(\mathbf{y})$, $\mathbf{y} - \eta\mathbf{e} \leq_C \mathbf{u}_j \leq_C s(\mathbf{y})$. For the upper estimate, the inclusion $B_\varphi(\mathbf{y}) \subseteq \Downarrow\mathbf{y}$ gives $s(\mathbf{y}) \in \Downarrow\mathbf{y}$, so $s(\mathbf{y}) \ll \mathbf{y}$ and therefore $s(\mathbf{y}) \leq_C \mathbf{y}$. We have therefore constructed a finite-image C-monotone map $s : P \to H$ satisfying the two inequalities in Proposition 4.3, contradicting Proposition 4.3.

Thus the maps $S_1, \ldots, S_m$, each way below Q_{Disc}, have no common upper bound in $\Downarrow Q_{\mathsf{Disc}}$. If $[\mathsf{Disc} \to E_{\mathsf{Disc}}]$ were continuous, then $\Downarrow Q_{\mathsf{Disc}}$ would be directed, and a finite subset of it would have a common upper bound in it. Hence the function space is not continuous. □

Corollary 4.5. *There is a pointed algebraic domain E such that $\mathsf{Disc} \in \mathbf{FS}$ and $[\mathsf{Disc} \to E]$ is not continuous.*

Proof. By Lemma 3.1, E_{Disc} is a pointed algebraic domain. The disk domain belongs to **FS** by the classical results cited at the beginning of this section, and Theorem 4.4 shows that $[\mathsf{Disc} \to E_{\mathsf{Disc}}]$ is not continuous. Thus one may take $E = E_{\mathsf{Disc}}$. □

The preceding counterexample concerns only continuity of the function space. If the function space is itself required to be an FS-domain, the situation is simpler and follows formally from Cartesian closure and retract closure.

Remark 4.6. For every domain D,

$$D \in \mathbf{FS} \quad \Longleftrightarrow \quad [D \to D] \in \mathbf{FS}.$$

Indeed, the forward implication follows from the Cartesian closure of **FS**. Conversely, fix $d_0 \in D$. The constant-map embedding $\mathrm{const} : D \to [D \to D]$ and the evaluation map $\mathrm{ev}_{d_0} : [D \to D] \to D$ satisfy $\mathrm{ev}_{d_0} \circ \mathrm{const} = \mathrm{id}_D$. Thus D is a Scott-continuous retract of $[D \to D]$, and the converse follows from the retract closure of FS-domains.

More generally, let S be any fixed FS-domain. For every domain E, the following conditions are equivalent:

$$E \in \mathbf{FS}, \qquad [S \to E] \in \mathbf{FS}, \qquad [D \to E] \in \mathbf{FS} \text{ for every } D \in \mathbf{FS}.$$

The only nonformal implication is again the retract argument. If $[S \to E]$ is an FS-domain, choose $s_0 \in S$. The constant-map embedding $E \to [S \to E]$ and evaluation at s_0 exhibit E as a Scott-continuous retract of $[S \to E]$.

If the output condition is weakened from being an FS-domain to being merely continuous, neither the source nor the target is characterized by membership in **FS**. The following standard examples illustrate the two failures. A dcpo L is $\ll$-separating [12] if it has an approximate identity $\mathcal{A}$ such that $\delta(x) \ll x$ for every $\delta \in \mathcal{A}$ and $x \in L$.

Remark 4.7. Consider the algebraic domain $\mathsf{Comb} = \{v_n, w_n : n \in \mathbb{N}\}$ ordered by

$$v_m \le v_n \quad \Longleftrightarrow \quad m \ge n, \qquad w_m \le v_n \quad \Longleftrightarrow \quad m \ge n,$$

where distinct w_m, w_n are incomparable and no v_m lies below a w_n. Every directed subset has a greatest element, so every element is compact and Comb is algebraic. The identity map shows that Comb is $\ll$-separating.

However, Comb is not an FS-domain. In fact, no map $f : \mathsf{Comb} \to \mathsf{Comb}$ is finitely separated from the identity. Suppose that a finite set $M \subseteq \mathsf{Comb}$ separates such a map from the identity, and choose N larger than every index occurring in an element of M. Applying finite separation to v_N would give some $m \in M$ such that $f(v_N) \le m \le v_N$. Every element below v_N is of the form v_k or w_k with $k \ge N$, but no such element belongs to M, a contradiction. Thus $\mathsf{Comb} \notin \mathbf{FS}$.

Nevertheless, Luan's theorem [12, Theorem 4.1] implies that

$$[D \to \mathsf{Comb}] \in \ll\mathbf{Sep}$$

for every FS-domain D, and hence $[D \to \mathsf{Comb}]$ is continuous. Thus the class of targets E for which $[D \to E]$ is continuous for every $D \in \mathbf{FS}$ is strictly larger than **FS**.

There is also a failure in the source variable. The standard Plotkin tie Tie is a pointed algebraic Noetherian domain which is not $\ll$-separating, and hence is not an FS-domain [12, Section 3]. By Luan's Noetherian-source theorem [12, Theorem 4.5],

$$[\mathsf{Tie} \to E] \in \ll\mathbf{Sep}$$

for every FS-domain E. Hence a domain outside **FS** may have continuous function spaces into every FS target.

The opposite extreme is provided by the disk-domain construction. Theorem 4.4 gives an FS-domain Disc and a pointed algebraic domain E_{Disc} such that $[\mathsf{Disc} \to E_{\mathsf{Disc}}]$ is not continuous. In particular, E_{Disc} is not $\ll$-separating. Otherwise Luan's theorem would imply that $[\mathsf{Disc} \to E_{\mathsf{Disc}}]$ is $\ll$-separating and hence continuous. Since every FS-domain is $\ll$-separating, it follows that $E_{\mathsf{Disc}} \notin \mathbf{FS}$.

These examples show that replacing the requirement $[D \to E] \in \mathbf{FS}$ by the weaker requirement that $[D \to E]$ be continuous destroys both the source-side and the target-side FS characterizations.

There remains a simple necessary condition on the source. If D is a domain such that $[D \to E]$ is continuous for every pointed FS-domain E, then D is coherent. Let $N_6 = \{\bot, a, b, c, d, \top\}$, where $\bot < a, b < c, d < \top$, the elements a, b are incomparable, the elements c, d are incomparable, and each of a, b lies below each of c, d. The finite domain N_6 is a pointed FS-domain but is not an L-domain. Hence $[D \to N_6]$ is continuous, and the bifurcation lemma [1, Lemma 4.3.1] forces D to be coherent.

The preceding results leave the following two natural questions open.

Problem 1. Let D be a coherent domain and let E be a pointed FS-domain. Is the function space $[D \to E]$ necessarily continuous?

The coherence assumption is necessary. Indeed, if $[D \to E]$ is continuous for every pointed FS-domain E, then taking $E = N_6$ and applying the bifurcation lemma shows that D must be coherent. Positive results are known when D is Noetherian, and when the target is restricted to pointed RB-domains, but neither result settles the question for arbitrary coherent sources and pointed FS targets.

Problem 2. Suppose that E is a domain such that $[D \to E]$ is continuous for every FS-domain D. Must E be a $\ll$-separating domain?

Every $\ll$-separating domain has this universal target property [13]. The converse would therefore give an intrinsic characterization of all targets that preserve continuity against arbitrary FS sources. The examples above show that this class is strictly larger than the class of FS-domains, but they do not determine whether it is strictly larger than the class of $\ll$-separating domains.

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